

18 The solubilities of BaSO_4 and PbSO_4 are x and $y \text{ mol dm}^{-3}$ respectively at 298 K.

A solution saturated with both BaSO_4 and PbSO_4 was prepared by dissolving $\text{BaSO}_4(\text{s})$ and $\text{PbSO}_4(\text{s})$ in water at 298 K.

Which statements about the saturated solution are correct?

1 $[\text{SO}_4^{2-}] = (x + y) \text{ mol dm}^{-3}$

2 $[\text{Ba}^{2+}] < x \text{ mol dm}^{-3}$

3 $[\text{SO}_4^{2-}] = [\text{Ba}^{2+}] + [\text{Pb}^{2+}]$

A 1, 2 and 3

B 1 and 2 only

C 1 and 3 only

D 2 and 3 only